

Quiz 1 Solutions

CPSC 406 — Algorithm Analysis, Section 1

Chapman University · Spring 2026

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1. (2 points) (a) The correct answer is

$$L_4 = \{ w \in \Sigma^* \mid |w|_a \equiv 0 \pmod{3} \}.$$

The DFA starts in the accepting state p , every a moves one state forward around the 3-cycle, and every b leaves the state unchanged. Therefore the machine accepts exactly when the number of a 's is divisible by 3.

The other choices are wrong because L_1 depends on total length rather than the number of a 's, L_2 allows exactly three a 's rather than any multiple of three, and L_3 counts both a 's and b 's even though b 's do not change the state. “None of these” is wrong because L_4 is correct.

- (b) The correct answer is

$$abc.$$

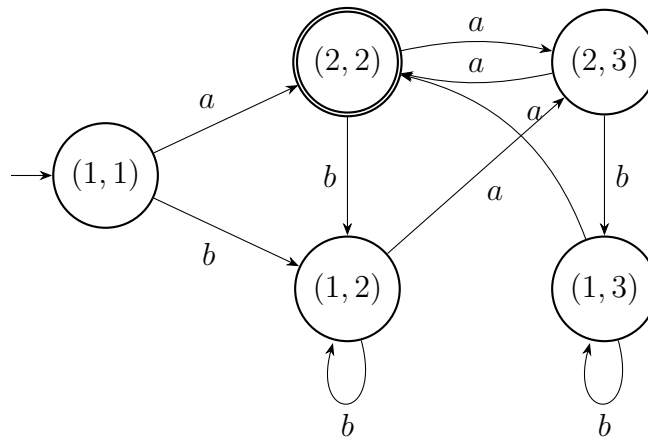
One accepting path is

$$1 \xrightarrow{a} 2 \xrightarrow{b} 1 \xrightarrow{c} 4.$$

Since state 4 is accepting, abc is accepted.

The word aab is not accepted: after $1 \xrightarrow{a} 2 \xrightarrow{a} 1$, there is no b -transition out of state 1. The word ccc is not accepted because the first c may move from 1 to accepting state 4, but there are still two symbols left and no outgoing transitions from 4 are shown. Thus “All of these” and “None of these” are both wrong.

2. (2 points) The standard product construction gives a DFA whose states are pairs (i, j) , where i is a state of $\mathcal{A}_1$ and j is a state of $\mathcal{A}_2$. A pair is accepting exactly when both components are accepting.



Only $(2, 2)$ is accepting because state 2 is accepting in both original DFAs.